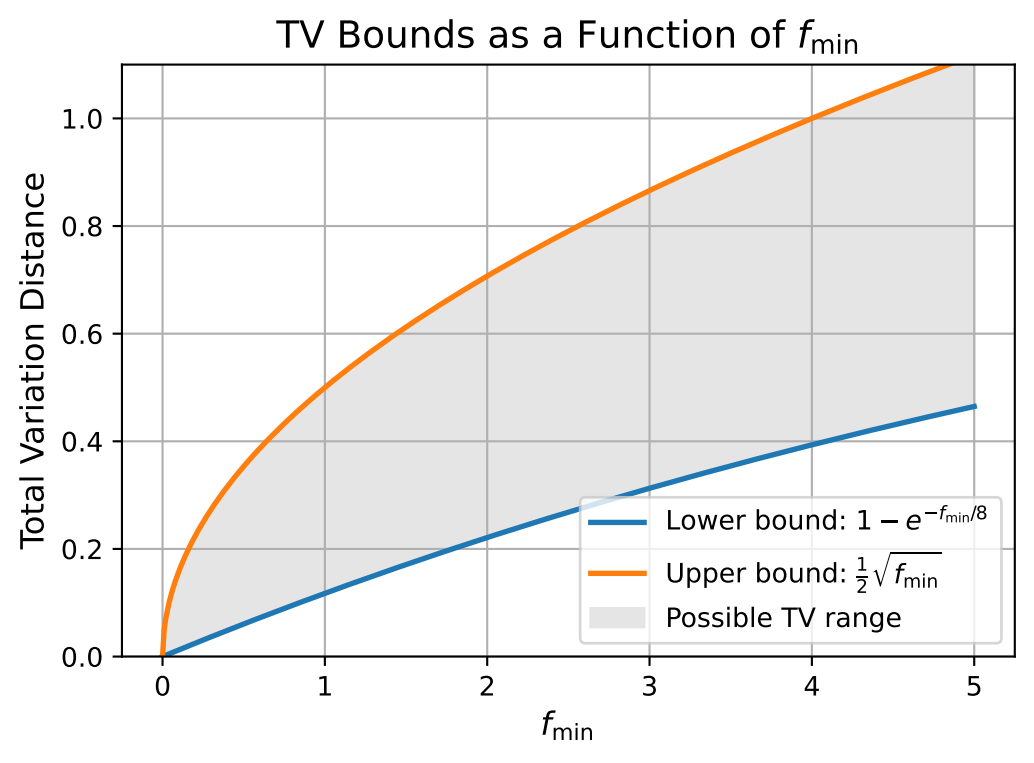
TV Bounds as a Function of $f_{\min}$
Total Variation Distance
$f_{\min}$
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
Lower bound: $1 - e^{-f_{\min}/8}$
Upper bound: $\frac{1}{2}\sqrt{f_{\min}}$
Possible TV range